Solution

Exponentials, Steps, Impulses, and System Properties

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1. The symbol $u(t)$ denotes a _____ signal.

   a) step

   b) impulse

   c) exponential

2. The symbol $\delta(t)$ denotes a _____ signal.

   a) step

   b) impulse

   c) exponential

3. The output of a causal system depends on _____ values of the input.

   a) past and present

   b) present and future

   c) present

4. A linear system satisfies the principle of _____.

   a) superposition

   b) stationarity

5. A signal $x(t) = e^{st}$ is a _____.

   a) discrete-time exponential

   b) continuous-time exponential

   c) continuous-time impulse

6. An exponential signal is defined as $x[n] = Az^n$. Which value for z results in the signal whose magnitude decays the fastest as n increases?

a) $z = 1.3$

b) $z = -1.1$

c) $z = 1$

d) $z = 0.9$

e) $z = -0.8$

f) $z = 0.7$

z^n decays fastest for $|z|$ smallest

Verify your answer using DTExponential.m.

7. A signal $x[n] = (0.9)^n \cos(\frac{\pi}{8}n)$. Write this signal as a sum of two exponentials, $x[n] = A_1 z_1^n + A_2 z_2^n$ and identify the correct values of A_1, z_1, A_2, z_2.

a) $A_1 = A_2 = 0.5; z_1 = 0.9, z_2 = -0.9$

b) $A_1 = A_2 = 0.5; z_1 = 0.9e^{j\pi/8}, z_2 = -0.9e^{j\pi/8}$

c) $A_1 = A_2 = 0.5; z_1 = 0.9e^{j\pi/8}, z_2 = 0.9e^{-j\pi/8}$

d) $A_1 = A_2 = 0.9; z_1 = j\pi/8, z_2 = -j\pi/8$

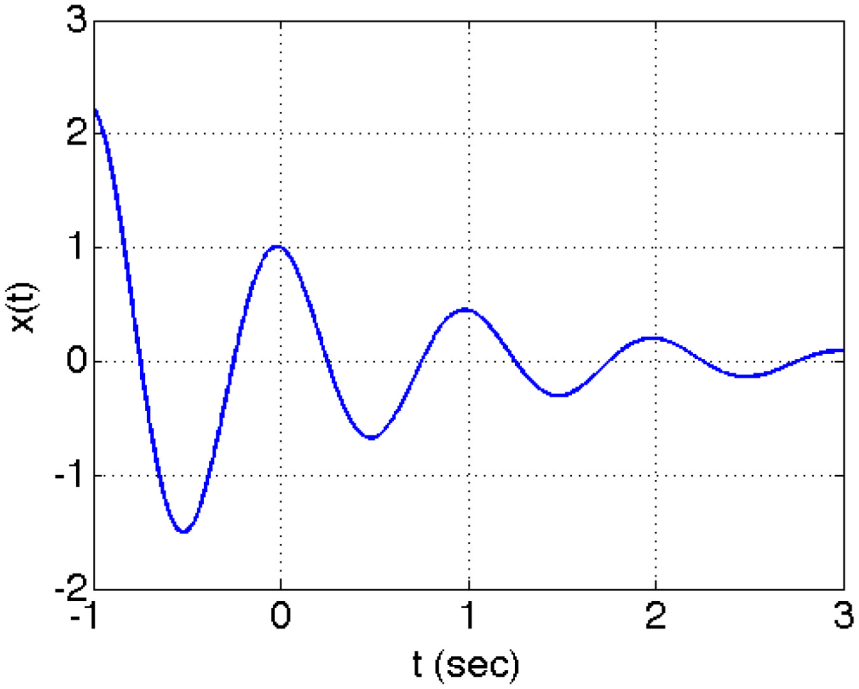
e) $A_1 = A_2 = 0.5; z_1 = 0.9 + j\pi/8, z_2 = 0.9 - j\pi/8$

$$\cos(\pi/8 n) = \frac{1}{2} e^{j\pi/8 n} + \frac{1}{2} e^{-j\pi/8 n}$$

so $$x[n] = \frac{1}{2} 0.9^n e^{j\pi/8 n} + \frac{1}{2} 0.9^n e^{-j\pi/8 n}$$

$$= \frac{1}{2} \left(0.9 e^{j\pi/8}\right)^n + \frac{1}{2} \left(0.9 e^{-j\frac{\pi}{8}}\right)^n$$

8. The signal depicted below is an exponentially-damped sinusoid of the form $x(t) = e^{at}\cos(2\pi ft)$. The value $x(1) = 0.4493$. Enter f and a.



Oscillating part repeats every 1 sec

$\Rightarrow f = 1$ Hz

$x(1) = e^{a}\cos(2\pi) = 0.4493$ (1)

$a = -0.8$

9. Which of the following choices describes the signal shown below?

a) $x(t) = 2u(t-1)$

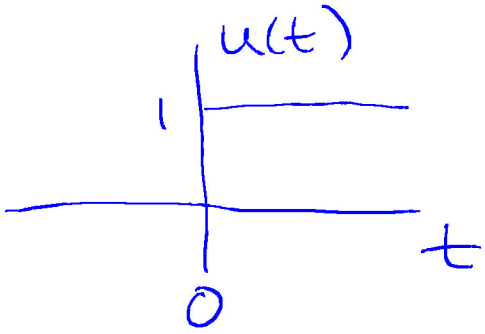
b) $x(t) = 2u(t-2)$

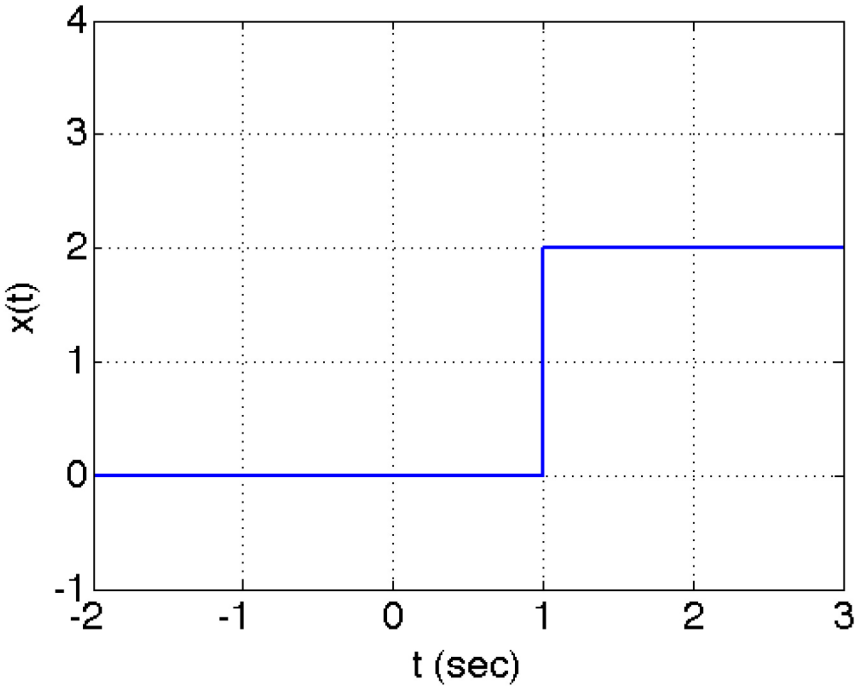
c) $x(t) = 2u(t+1)$

d) $x(t) = 3u(t-1)$

e) $x(t) = 3u(t-2)$

f) $x(t) = 3u(t+1)$





shift u(t) 1 sec to the right

scale amplitude by 2

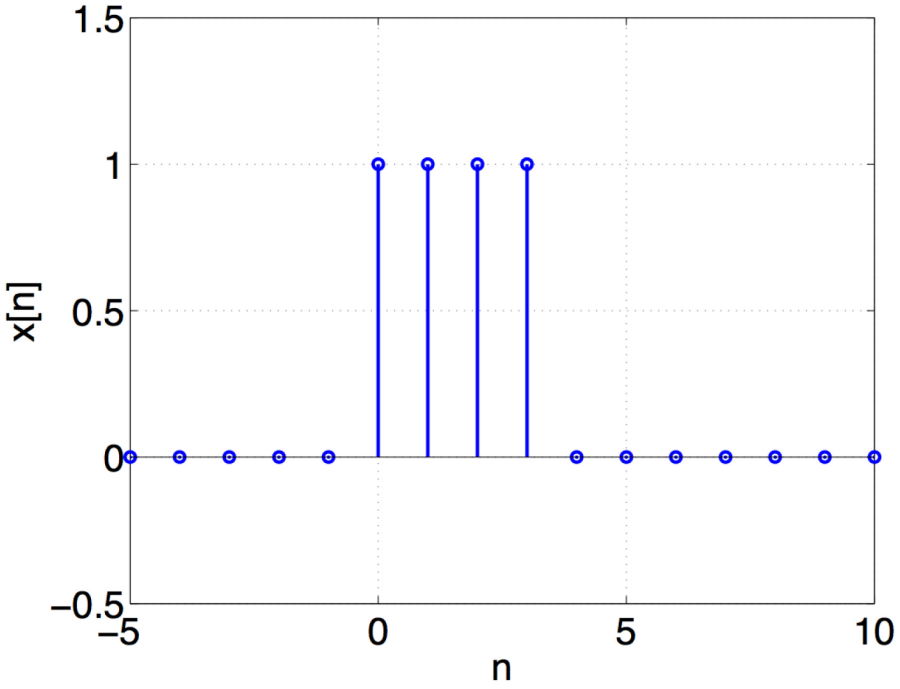
10. Which of the following choices describes the signal shown below?

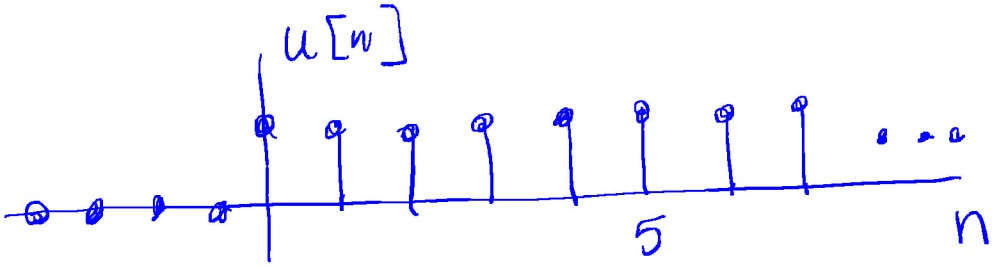
a) $x[n] = u[n] - u[n-4]$

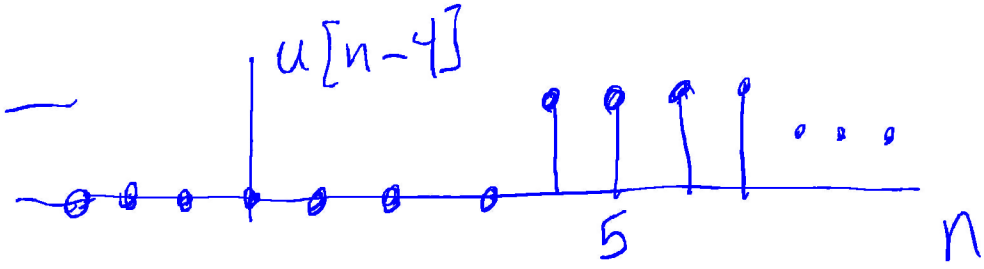
b) $x[n] = u[n-1] - u[n-5]$

c) $x[n] = u[n] - u[n-6]$

d) $x[n] = u[n+1] - u[n-3]$







11. Use the sifting property of the impulse to evaluate

$$b = \int_{-\infty}^{\infty} x(t)\delta(t-a)dt$$

for $x(t) = 2e^{-t}\cos(\pi t)$ if $a = 1$.

Sifting property: $\int_{-\infty}^{\infty} x(t)\,\delta(t-a)dt = x(a)$

Since $a=1$, $x(t) = 2e^{-t}\cos(\pi t)$

$$x(a) = 2e^{-1}\cos(\pi)$$

$$= -2e^{-1}$$

Questions 12-14 apply to the systems with input $x[n]$ and output $y[n]$ defined by the following equations.

a) $y[n] = (x[n])^2 - x[n+1]$

b) $y[n] = 2x[-n] + x[n-1]$

c) $y[n] = \cos(x[n-1])$

d) $y[n] = n^2 x[n]$

output only depends on past and/or present inputs

12. Which of the systems are causal?

c, d are causal

a is not since $y[n]$ depends on $x[n+1]$

b is not since $y[-1]$ depends on $x[1]$

13. Which of the systems are linear?

linear system: if $y_1[n] = H\{x_1[n]\}$
$y_2[n] = H\{x_2[n]\}$

then $y[n] = H\{a x_1[n] + b x_2[n]\} = a y_1[n] + b y_2[n]$

a) $y[n] = (a x_1[n] + b x_2[n])^2 - a x_1[n+1] - b x_2[n+1]$

$= a^2 x_1^2[n] + 2ab x_1[n] x_2[n] + b^2 x_2^2[n]$
$- a x_1[n+1] - b x_2[n+1]$

not linear

$\neq a x_1^2[n] - a x_1[n+1] + b x_2^2[n] - b x_2[n+1]$

b) $y[n] = 2(a x_1[-n] + b x_2[-n]) + a x_1[n-1] + b x_2[n-1]$

linear $= a(2x_1[-n] + x_1[n-1]) + b(2x_2[n] + b x_2[n-1])$

c) $\cos(a x_1[n-1] + b x_2[n-1]) \neq a\cos(x_1[n-1])$
$+ b\cos(x_2[n-1])$

not linear

d) $y[n] = n^2(a x_1[n] + b x_2[n])$

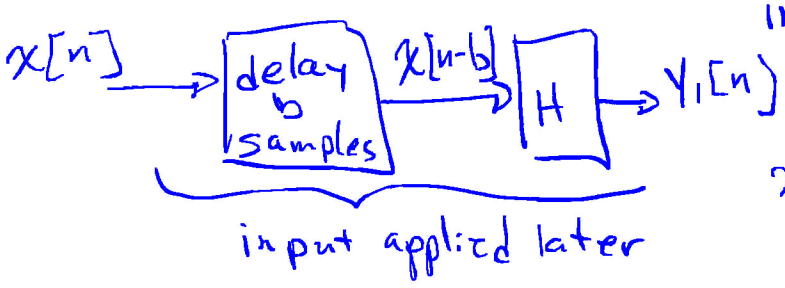
$= a\, n^2 x_1[n] + b\, n^2 x_2[n]$

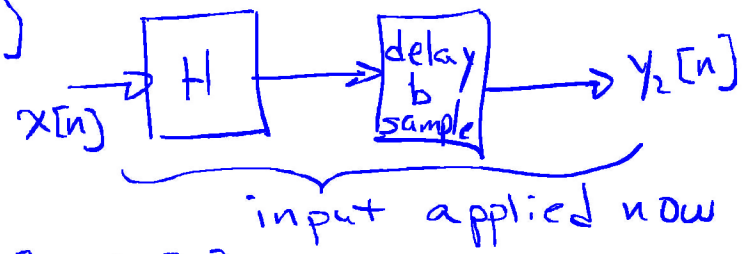
$= a\, y_1[n] + b\, y_2[n]$

linear

14. Which of the systems are time-invariant?

output is same whether input is applied now or later





$H\{\cdot\}$ time invariant iff $y_1[n] = y_2[n]$

a) $y_1[n] = (x[n-b])^2 - x[n-b+1] = y_2[n]$

time invariant

b) $y_1[n] = 2x[-(n-b)] + x[n-b-1] = 2x[-n+b] + x[n-b-1]$

$y_2[n] = 2x[-n-b] + x[n-b-1]$ time varying

$= \text{Delay by } b \{ 2x[-n] + x[n-1] \}$

c) $y_1[n] = \cos(x[n-b-1])$ $y_2[n] = \text{delay by } b \{\cos(x[n-1])\}$

time invariant $= \cos(x[n-b-1])$

d) $y_1[n] = n^2 x[n-b]$ $y_2[n] = \text{delay by } b \{ n^2 x[n] \}$

$= (n-b)^2 x[n-b]$

time varying

Problems 15-19 explore the effect of two different moving average systems.

System 1 evaluates the output $y_1[n]$ from the input $x[n]$ as

$$y_1[n] = b_0x[n] + b_1x[n-1] + b_2x[n-2] + b_3x[n-3]$$

where $b_0 = 0.25$; $b_1 = 0.25$; $b_2 = 0.25$; $b_3 = 0.25$

System 2 is of the same form but with different coefficients

$$y_2[n] = c_0x[n] + c_1x[n-1] + c_2x[n-2] + c_3x[n-3]$$

The c_k are $c_0 = -0.5$; $c_1 = 0.5$; $c_2 = 0$; $c_3 = 0$

15. System 1 forms the _____ input values.

a) average of the 2 most recent

b) average of the 3 most recent

c) average of the 4 most recent

d) difference of the 2 most recent

e) difference of the 4 most recent

16. Therefore system 1 will _____ rapid fluctuations or high-frequency components of the input signal.

a) smooth out/attenuate

b) enhance/amplify

average over 4 samples ⇒ smooth fluctuations over those 4

17. System 2 forms the _____ input values.

a) average of the 2 most recent

b) difference of the 2 most recent

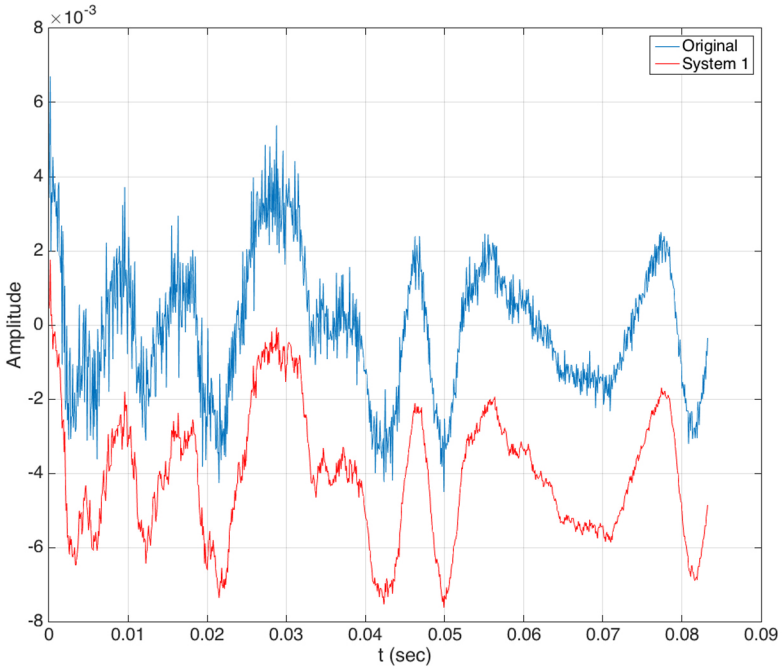
18. Therefore system 2 will attenuate _____ components of the input.

a) low-frequency/slow variation

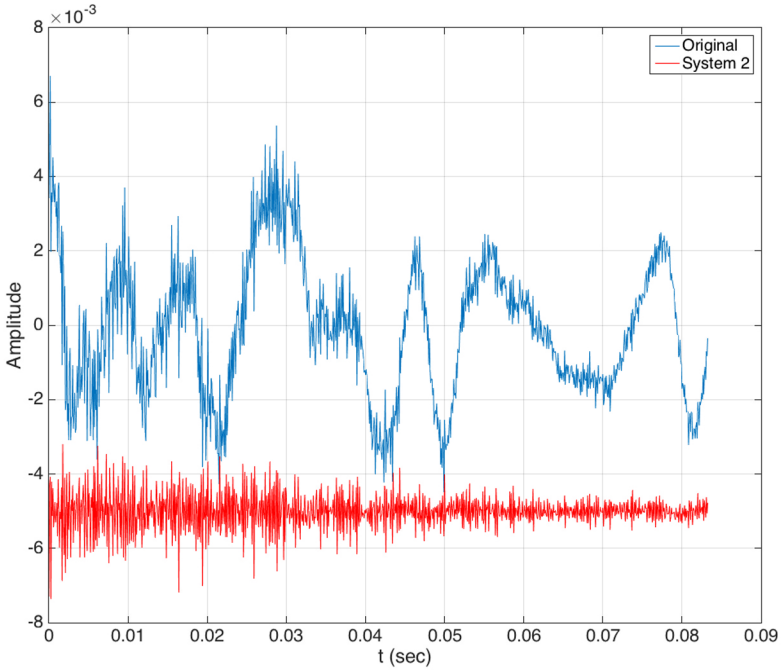
b) high-frequency/fast variation

taking difference removes the part that is common to those two values

19. Download the files SystemIntro.m and StealthGroover.wav, enter the values for b_k and c_k given prior to Problem 15 in lines 18 and 21 of SystemIntro.m. Then run the code to compare the sound of the original music clip to those processed as described prior to Problem 15. Note that the `pause` command is used after playing the audio clips to keep them from being played simultaneously. You need to hit a key in the MATLAB command window to resume the code after each audio clip. Also, the plots show a short section (about 80 ms) of the input (blue) and output (red) waveforms where the output has been offset vertically for visibility.



fast fluctuations removed, sounds somewhat dull due to reduced high frequency content



slow variations removed, sounds very bright due to reduced low frequency content